UF UNIVERSITY of FLORIDA

Department of Mathematics

MAP 2302
Exam 3A
Spring 2023

A. Sign your bubble sheet on the back at the bottom in ink.

B. In pencil, write and encode in the spaces indicated (-5 for errors):

1) Name (last name, first initial, middle initial)

2) UF ID number

3) Section number

C. Under "special codes" code in the test ID numbers 3, 1.

1	2	●	4	5	6	7	8	9	0
●	2	3	4	5	6	7	8	9	0

D. At the top right of your answer sheet, for "Test Form Code", encode A. (-5 for errors)

●	B	C	D	E

E. 1) This test consists of 12 multiple choice questions of 5.5 points in value, plus 4 free response questions worth 40 points.

2) The time allowed is 100 minutes.

3) You may write on the test.

4) Raise your hand if you need more scratch paper or if you have a problem with your test. DO NOT LEAVE YOUR SEAT UNLESS YOU ARE FINISHED WITH THE TEST.

F. KEEP YOUR BUBBLE SHEET COVERED AT ALL TIMES.

G. When you are finished:

1) Before turning in your test **check carefully for transcribing errors**. Any mistakes you leave in are there to stay. Blank scantrons will count as zero and are not grounds for a retest.

2) You must turn in your scantron and tearoff sheets to your discussion leader or exam proctor. Be prepared to show your picture I.D. with a legible signature.

3) The answers will be posted in Canvas within one day after the exam. Your instructor will return your tearoff sheet in class. Your score will also be posted in Canvas within one week of the exam.

NOTE: Be sure to bubble the answers to questions 1−12 on your scantron.

Questions 1 − 12 are worth 5.5 points each.

1. Which of the following functions is of exponential order?

A. $f(x) = e^{x^3}$
B. $f(x) = e^{e^x}$
C. $f(x) = e^{x^2}\sin(e^{x^2})$
D. $f(x) = e^{x^3}\cos(e^{x^2})$
E. $f(x) = e^x\sin(e^{x^2})$

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2. Find the Laplace transform $\mathscr{L}\{t\delta(t-1)\}$.

A. $F(s) = 1$
B. $F(s) = e^{2s}$
C. $F(s) = se^{-s}$
D. $F(s) = -e^{-s}$
E. $F(s) = e^{-s}$

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3. Which statement is correct about the following piecewise function?

$$f(x) = \begin{cases} x+1 & x < 0 \\ x^2+1 & 0 \le x < 2 \\ \frac{1}{x-1} & x \ge 2 \end{cases}$$

A. $f(x)$ is continuous on $(-\infty, \infty)$
B. $f(x)$ is continuous on $[0, 3]$
C. $f(x)$ is piecewise continuous on $(-\infty, \infty)$
D. $f(x)$ is neither piecewise continuous nor continuous on $[-1, 1]$
E. $f(x)$ is neither piecewise continuous nor continuous on $[1, 3]$

4. Solve the initial value problem.

$$y'' + y = \delta(t - 2\pi),\ y(0) = 0,\ y'(0) = 1$$

A. $\sin t + u(t - 2\pi)\sin t$
B. $\cos t + u(t - 2\pi)\cos t$
C. $\sin t + u(t - 2\pi)\cos t$
D. $\cos t + u(t - 2\pi)\sin t$
E. $[t + u(t - 2\pi)]\sin t$

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5. Use convolution to obtain a formula for the solution to the given initial value problem where g is piecewise continuous on $[0, \infty)$ and is of exponential order.

$$y'' + 6y' + 13y = g(t); \quad y(0) = 0, \quad y'(0) = 1$$

A. $\frac{1}{3}e^{-3t}\sin(2t) + \frac{1}{2}\int_0^t e^{-3(t-v)}\sin[2(t-v)]g(v)\,dv$

B. $\frac{1}{2}e^{-3t}\cos(2t) + \frac{1}{2}\int_0^t e^{-2(t-v)}\sin[3(t-v)]g(v)\,dv$

C. $\frac{1}{2}e^{-3t}\sin(2t) + \frac{1}{2}\int_0^t e^{-3(t-v)}\sin[2(t-v)]g(v)\,dv$

D. $\frac{1}{2}e^{-3t}\sin(2t) + \frac{1}{2}\int_0^t e^{2(t-v)}\cos[3(t-v)]g(v)\,dv$

E. $\frac{1}{2}e^{3t}\cos(2t) + \frac{1}{2}\int_0^t e^{3(t-v)}\cos[2(t-v)]g(v)\,dv$

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6. Solve the integral equation

$$y(t) = 1 + \int_0^t y(v)(t - v)dv$$

.

A. $\frac{1}{2}(e^t - e^{-t})$ B. $\cos t$ C. $1 - \frac{t^2}{2}$

D. $\frac{1}{2}(e^t + e^{-t})$ E. $\frac{1}{2}(e^t + te^t)$

7. Determine

$$\mathscr{L}^{-1}\left\{\frac{5s^2-3s+6}{(s-1)(s^2+2s+5)}\right\}$$

.

A. $4e^t\cos(2t)-\frac{5}{2}e^t\sin(2t)+e^t$

B. $4e^{-t}\cos(t)-\frac{5}{2}e^{-t}\sin(t)+e^t$

C. $4e^{-t}\cos(2t)-\frac{5}{2}e^{-t}\sin(2t)+e^t$

D. $4e^t\cos(-t)-\frac{5}{2}e^t\sin(-t)+e^t$

E. $4e^{-t}\cos(2t)+\frac{5}{2}e^{-t}\sin(2t)+e^{-t}$

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8. Find $Y(s)$, the Laplace transform of the given initial value problem

$$y''+6y'+13y=5te^{2t}$$

with $y(0)=0, y'(0)=-1$.

A. $\dfrac{1-s^2-4s}{(s-2)^2(s^2+6s+13)}$

B. $\dfrac{5}{(s+2)^2(s^2+6s+13)}$

C. $\dfrac{s^2+4s-1}{(s-2)(s^2-6s+13)}$

D. $\dfrac{5}{(s-2)^2(s^2+6s+13)}$

E. $\dfrac{1-s^2+4s}{(s-2)^2(s^2+6s+13)}$

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9. Find $\mathscr{L}^{-1}\left\{\dfrac{e^{-5s}(5s^2+3s+4)}{(s+3)(s^2+1)}\right\}$.

A. $u(t-5)[4e^{-3t}+\cos(t)]$

B. $4e^{-3(t-5)}-4\cos(t-5)$

C. $4e^{-3(t-5)}-4\sin(t-5)$

D. $u(t-5)[4e^{-3(t-5)}+\sin(t-5)]$

E. $u(t-5)[4e^{-3(t-5)}+\cos(t-5)]$

10. Determine the Laplace transform of $g(t)$ using unit step functions where

$$g(t) = \begin{cases} -3 & \text{if } t < 2 \\ t^2 + 2 & \text{if } t \geq 2 \end{cases} \tag{1}$$

.

A. $e^{-2s}(\frac{2}{s^3} + \frac{4}{s^2} + \frac{9}{s})$

B. $-\frac{3}{s} + e^{-2s}(\frac{2}{s^3} + \frac{4}{s^2} + \frac{9}{s})$

C. $-\frac{3}{s} + e^{-2s}(\frac{5}{s^3} + \frac{9}{s})$

D. $e^{2s}(\frac{2}{s^3} + \frac{4}{s^2} + \frac{9}{s})$

E. $e^{2s}(\frac{5}{s^3} + \frac{9}{s})$

11., 12. A linear system is governed by the given initial value problem

$$y'' + 4y' - 21y = g(t),\ y(0) = 0,\ y'(0) = 10.$$

11. Find the transfer function $H(s)$.

A. $\frac{1}{(s-3)(s+7)}$

B. $\frac{G(s)}{(s+2)^2 - 28}$

C. $\frac{s}{(s-3)(s+7)} + \frac{G(s)}{(s-3)(s+7)}$

D. $\frac{s-3}{(s+2)^2 - 28} + \frac{s+2}{(s+2)^2 - 28}$

E. $\frac{s}{(s+7)} + \frac{G(s)}{(s-3)(s+7)}$

12. Find the impulse response function $h(t)$.

A. $\frac{e^{3t} - e^{-7t}}{10}$

B. $e^{-2t} \sin \sqrt{28t}$

C. $e^{-2t} \sin \sqrt{28t}$

D. $e^{-2t} \cos t + \int_0^t e^{-2t+2v} \sin(-2t + 2v) g(v)\, dv$

E. $e^{-2t} \sin t + \int_0^t e^{-2t+2v} \sin(-2t + 2v) g(v)\, dv$

Elementary Laplace Transforms

$f(t)$	$\mathcal{L}\{f(t)\}$
C	$\frac{C}{s}, s > 0$
t^n	$\frac{n!}{s^{n+1}}, s > 0$
e^{at}	$\frac{1}{s-a}, \quad s > a$
$e^{at}t^n$	$\frac{n!}{(s-a)^{n+1}}, \quad s > a,\ n = 0, 1, 2, \ldots$
$\sin(bt)$	$\frac{b}{s^2+b^2}, s > 0$
$\cos(bt)$	$\frac{s}{s^2+b^2}, s > 0$
$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}, \quad s > a$
$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}, \quad s > a$
$\sinh(bt)$	$\frac{b}{s^2-b^2}, s > \lvert b\rvert$
$\cosh(bt)$	$\frac{s}{s^2-b^2}, s > \lvert b\rvert$
$u(t-a)$	$\frac{e^{-as}}{s}, s > 0$
$\delta(t-a)$	e^{-as}

Properties of the Laplace transform

$\mathcal{L}\{f+g\} = \mathcal{L}\{f\} + \mathcal{L}\{g\}$

$\mathcal{L}\{cf\} = c\mathcal{L}\{f\}$ for any constant c

$\mathcal{L}\{e^{at}f(t)\}(s) = F(s-a)$ where $F(s) = \mathcal{L}\{f\}(s)$

$\mathcal{L}\{f'\}(s) = sF(s) - f(0)$

$\mathcal{L}\{f''\}(s) = s^2F(s) - sf(0) - f'(0)$

$\mathcal{L}\{f^{(n)}\}(s) = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - f^{(n-1)}(0)$

$\mathcal{L}\{t^nf(t)\} = (-1)^n\frac{d^nF}{ds^n}(s)$

$\mathcal{L}\{ty'\}(s) = -sY'(s) - Y(s)$ where $Y(s) = \mathcal{L}\{y\}(s)$

$\mathcal{L}\{ty''\}(s) = -s^2Y'(s) - 2sY(s) + y(0)$

MAP 2302 Exam 3A, Part II Free Response

Name (Last, First, legibly): _____ Section #: _____

(-5 for errors in name or section number)

SHOW ALL WORK TO RECEIVE FULL CREDIT

1. (8 pts) Find $\mathscr{L}\{t\cos(bt)\}$.

2. (10 pts) Solve the initial value problem

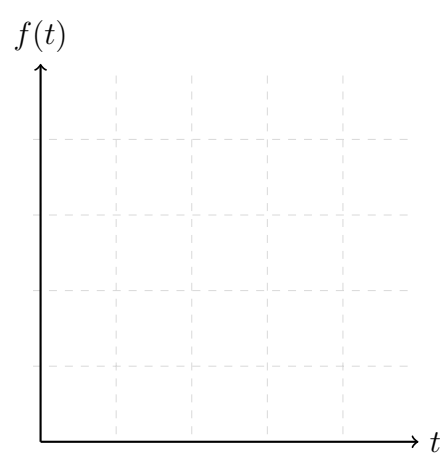
$$y'' + 4y' + 4y = u(t - \pi),\ y(0) = 0,\ y'(0) = 0.$$

3. (10 pts)

a. Find the Laplace transform of $f(t) = \int_0^t (t-v)e^{3v}\,dv$. (2 pts)

b. Evaluate $\int_{-\infty}^{\infty} (t^2-1)\delta(t)\,dt$. (2 pts)

c. Determine $\mathscr{L}^{-1}\left\{\frac{e^{-2s}}{s^2}\right\}$ and sketch its graph for $t \geq 0$. (6 pts)



4. (12 pts) Find the Laplace transform of $g(t)$ **using the definition of the Laplace transform. No credit will be granted for other methods or use of tables**.

$$g(t) = \begin{cases} t+2 & t \leq 4 \\ 3 & t > 4 \end{cases}$$

University of Florida Honor Pledge:

On my honor, I have neither given nor received unauthorized aid doing this exam.

Signature: _____